

relatorio

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Estimadores de Mínimos Quadrados

seja X a matriz de dados com n observações e p variáveis;

seja y a variável alvo

Os estimadores de mínimos quadrados são:

$$\hat{B} = (X'X)^{-1} X'y$$

Assumindo o modelo linear $Y = XB + e$,

e assumindo que, para todo i e j :

$$E(e_i) = 0$$

$$\text{cov}(e_i, e_j) = 0 \text{ (erros não correlacionados)}$$

$$\text{var}(e_i) = \sigma^2 \text{ (homocedasticidade)}$$

de acordo com o teorema de Gauss-Markov, os estimadores de mínimos quadrados serão os melhores estimadores lineares não viesados, sob as suposições anteriores.

Para os testes de hipóteses, assumimos que $e_i \sim N(0, \sigma^2)$

Imports

```
#install.packages("plotly")  
library("plotly")
```

```
source("functions.R")
```

Função Gaussiana

```
$$  
f(x, y) = K \cdot e^{-c(x^2 + y^2)}  
$$
```

$$f(x, y) = k \cdot \exp[-c(x^2 + y^2)]; k, c > 0$$

k : Amplitude máxima da função; $f(0,0) = k$

c : Abertura e decaimento da curva;

$f: \mathbb{R}^2 \rightarrow \mathbb{R}^+$

```
f <- function(x,y, k=10, c=6){
  return( k*exp(-c(x^2 + y^2)))
}
```

Objetivo

Usar polinômios de duas variáveis para aproximar a função gaussiana;

Geração de pontos aleatórios

```
set.seed(28)

# número de pontos
n<- 100

min_<- -4
max_<- 4

x_unif<- runif(n, min= min_, max= max_)
y_unif<- runif(n, min= min_, max= max_)
z_unif<- f(x_unif,y_unif)

x_normal<- rnorm(n, 0, 1)
y_normal<- rnorm(n, 0, 1)
z_normal<- f(x_normal, y_normal)

pontos_unif<- cbind(x_unif,y_unif, z_unif)
pontos_normal<- cbind(x_normal, y_normal, z_normal)

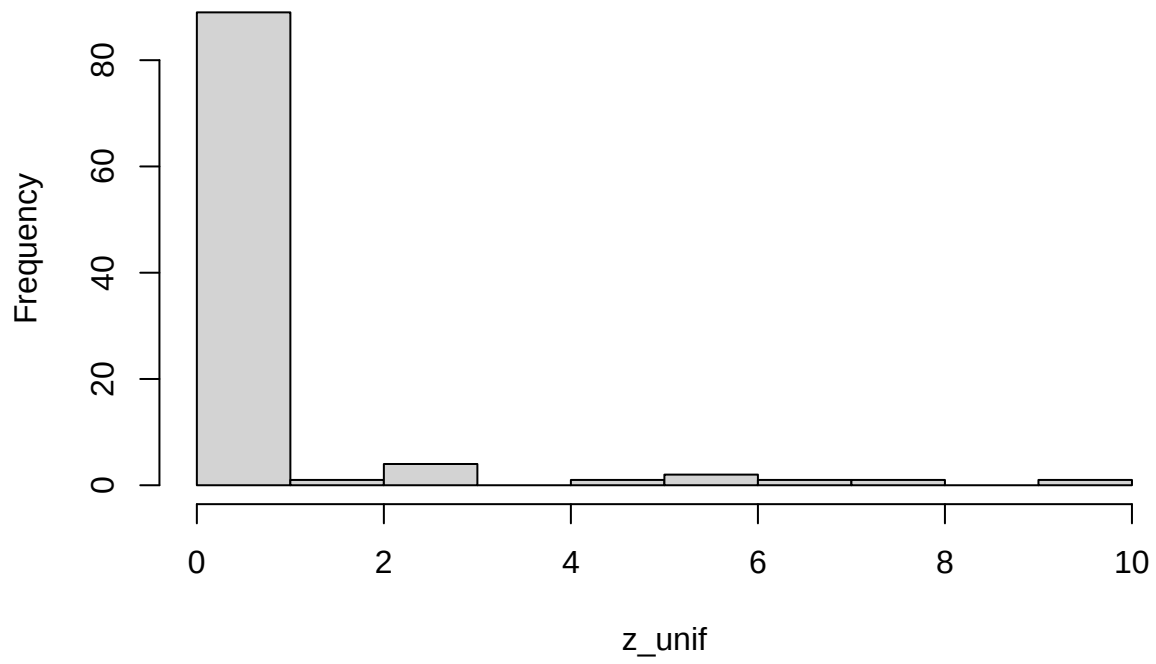
matriz_unif<- data.frame(pontos_unif)
matriz_normal<- data.frame(pontos_normal)

print(matriz_unif[1:5,])

##      x_unif    y_unif      z_unif
## 1 -3.771398  1.964150 1.404043e-07
## 2 -3.293045 -1.324710 3.375378e-05
## 3 -0.205058 -2.320177 4.403878e-02
## 4  3.083021  3.817274 3.496746e-10
## 5 -3.267464  1.095955 6.945364e-05

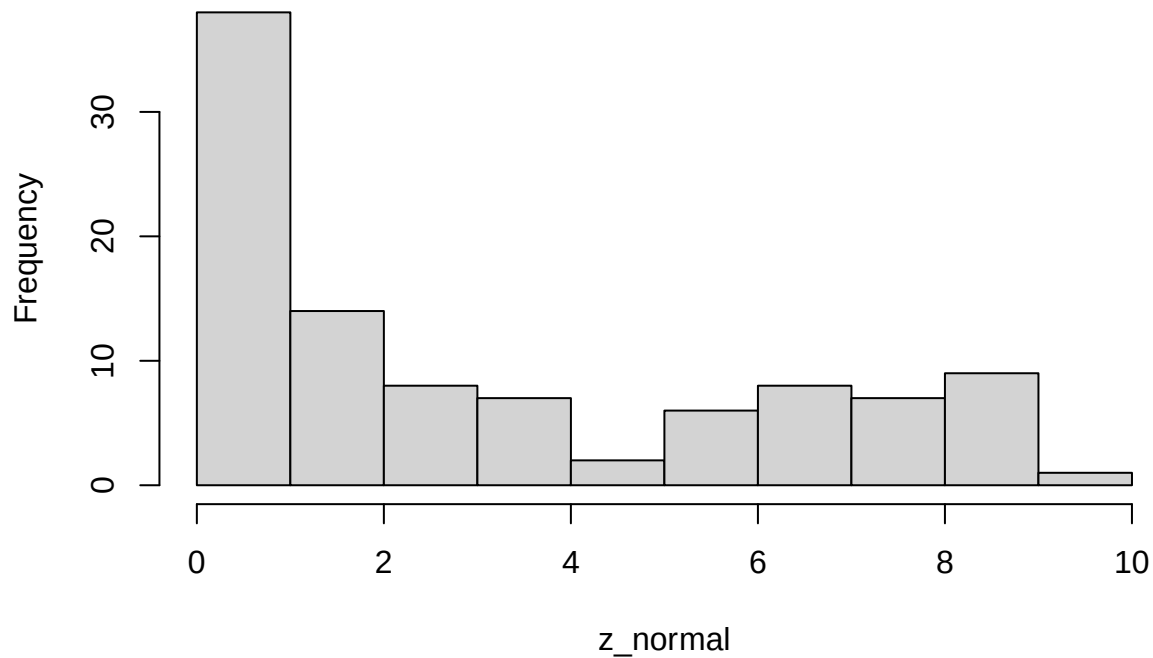
hist(z_unif)
```

Histogram of z_unif



```
hist(z_normal)
```

Histogram of z_normal

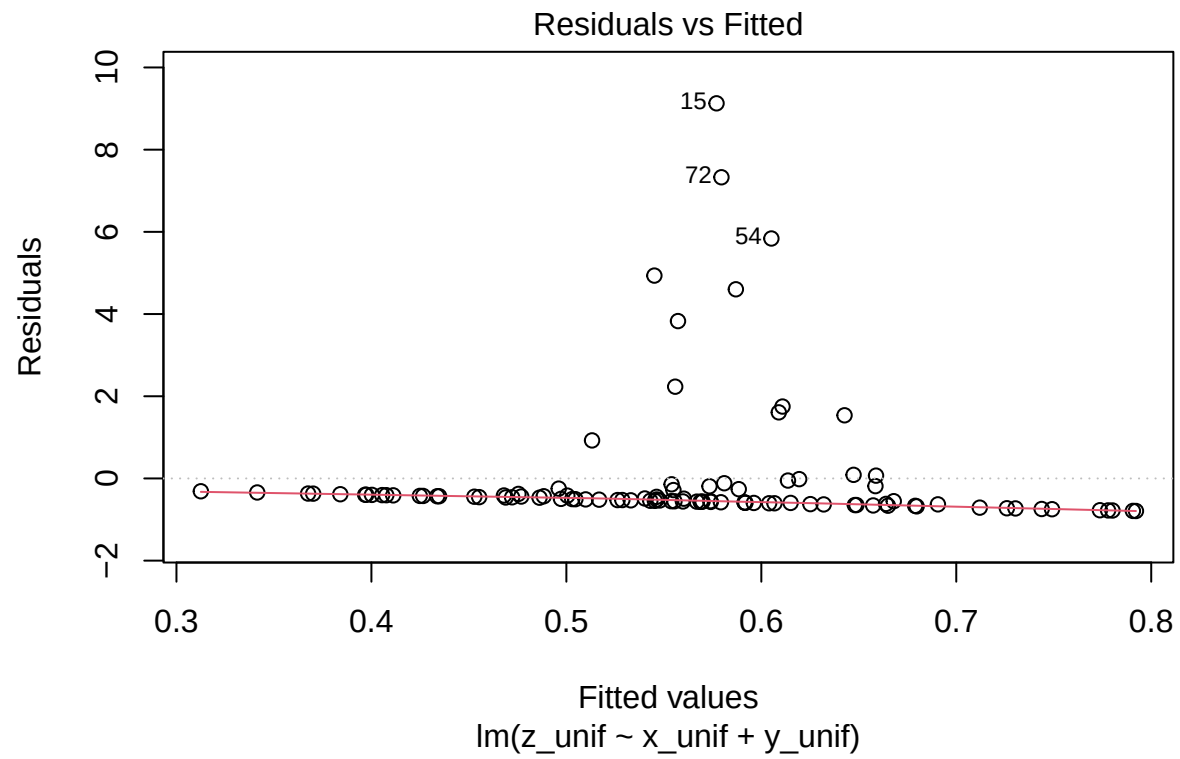


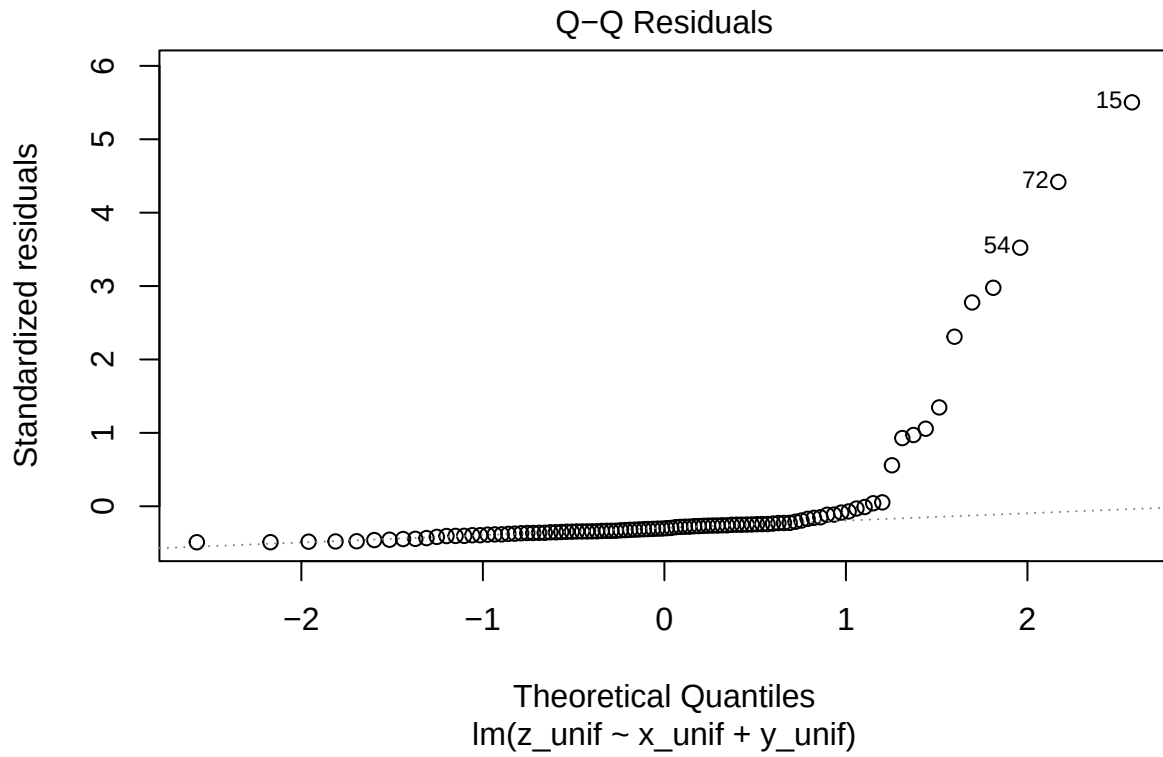
Modelo 1 [$m_1(x,y) = b_0 + b_1x + b_2y$]

```
modelo1 <- lm(z_unif ~ x_unif + y_unif, data=matriz_unif)
summary(modelo1)
```

```
##
## Call:
## lm(formula = z_unif ~ x_unif + y_unif, data = matriz_unif)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.7922 -0.5916 -0.4951 -0.3695  9.1272
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.58110    0.16942   3.43 0.000888 ***
## x_unif         0.03216    0.07472   0.43 0.667805
## y_unif        -0.03856    0.07274  -0.53 0.597208
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.667 on 97 degrees of freedom
## Multiple R-squared:  0.004074,    Adjusted R-squared:  -0.01646
## F-statistic: 0.1984 on 2 and 97 DF,  p-value: 0.8204
```

```
plot(modelo1, which = 1:2)
```



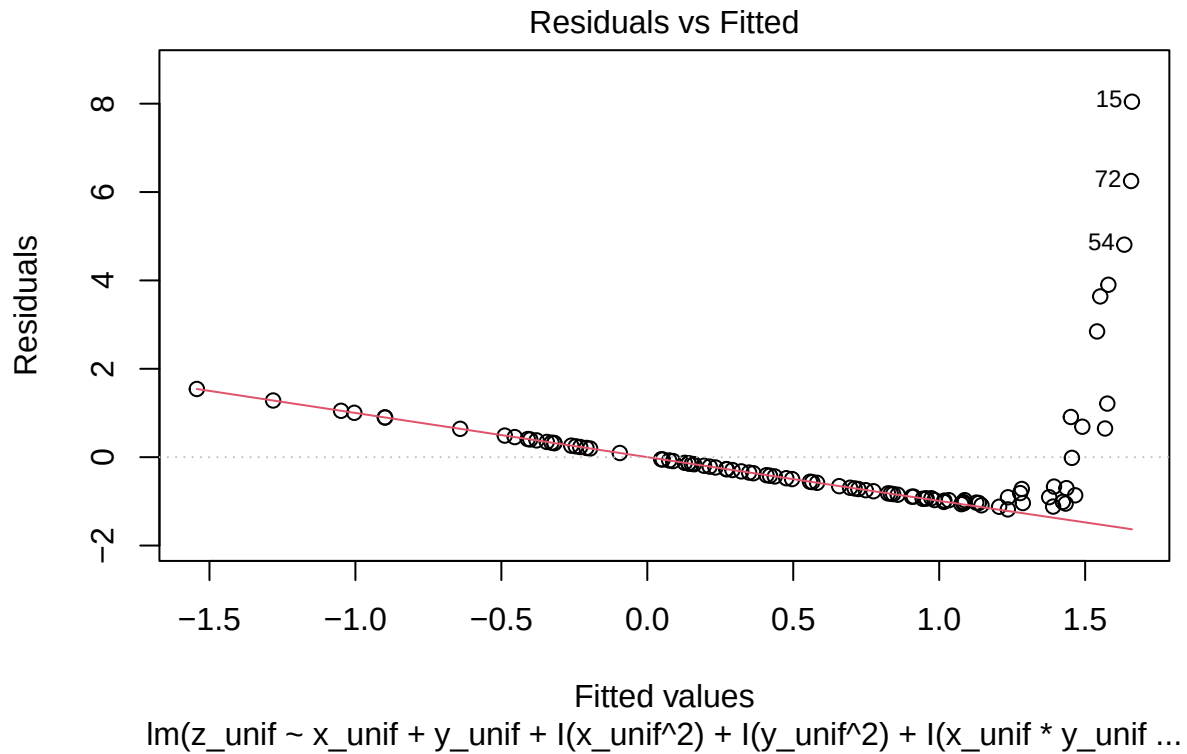


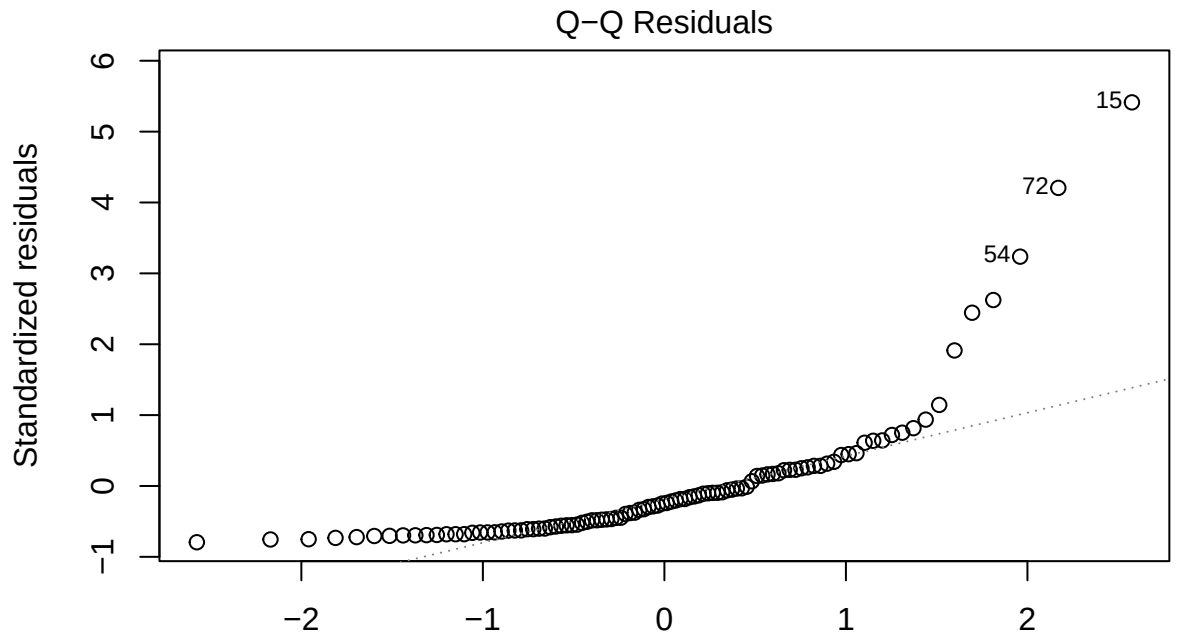
Modelo 2 [$m_2(x,y) = b_0 + b_1 \cdot x + b_2 \cdot y + b_3 \cdot x^2 + b_4 \cdot y^2 + b_5 \cdot x \cdot y$]

```
modelo2<- lm(z_unif ~ x_unif + y_unif + I(x_unif^2) + I(y_unif^2) + I(x_unif*y_unif), data=matriz_unif)
summary(modelo2)
```

```
##
## Call:
## lm(formula = z_unif ~ x_unif + y_unif + I(x_unif^2) + I(y_unif^2) +
##      I(x_unif * y_unif), data = matriz_unif)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.1823 -0.8937 -0.3560  0.3185  8.0439
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.65558    0.26678   6.206 1.46e-08 ***
## x_unif          0.03650    0.06990   0.522 0.602713
## y_unif          0.03422    0.06789   0.504 0.615417
## I(x_unif^2)    -0.09580    0.03255  -2.943 0.004094 **
## I(y_unif^2)    -0.11118    0.03266  -3.404 0.000979 ***
## I(x_unif * y_unif) 0.01511    0.02811   0.538 0.592143
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.51 on 94 degrees of freedom
## Multiple R-squared:  0.2083, Adjusted R-squared:  0.1662
## F-statistic: 4.947 on 5 and 94 DF,  p-value: 0.0004591
plot(modelo2, which = 1:2)
```





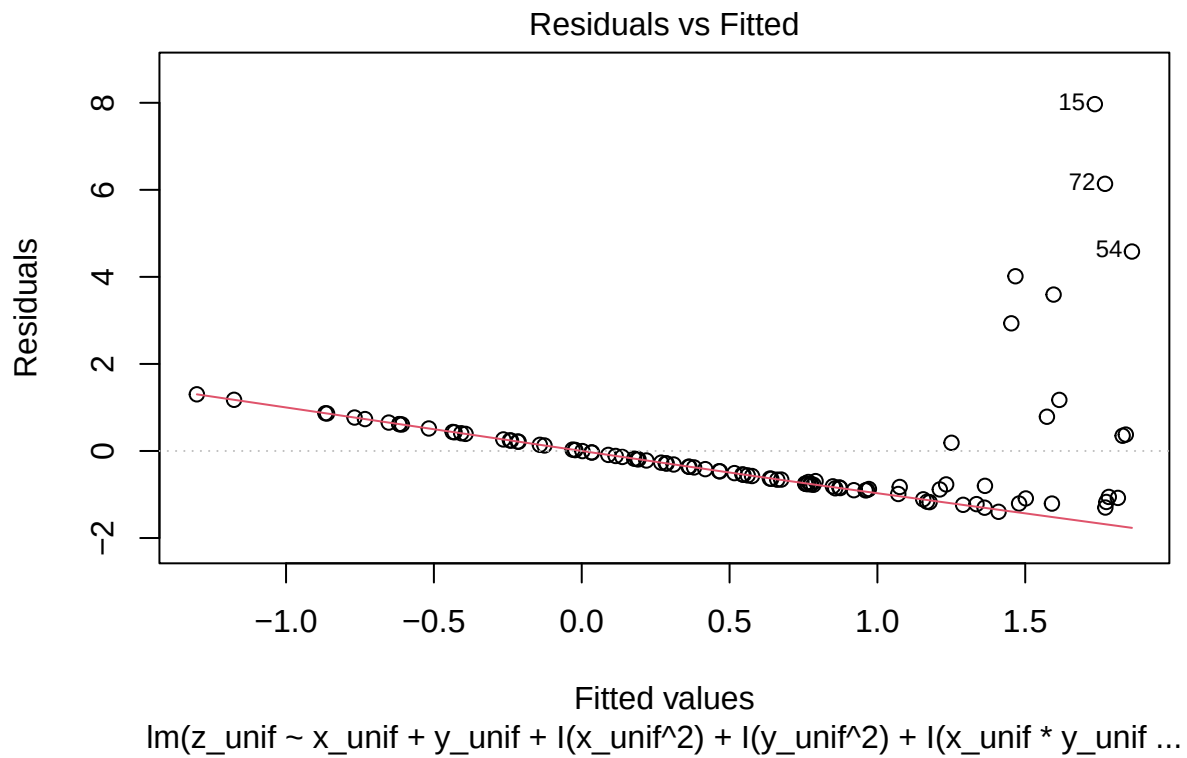
Modelo 3 [$m_2(x,y) = b_0 + b_1 \cdot x + b_2 \cdot y + b_3 \cdot x^2 + b_4 \cdot y^2 + b_5 \cdot x \cdot y + b_6 \cdot x^3 + b_7 \cdot y^3$]

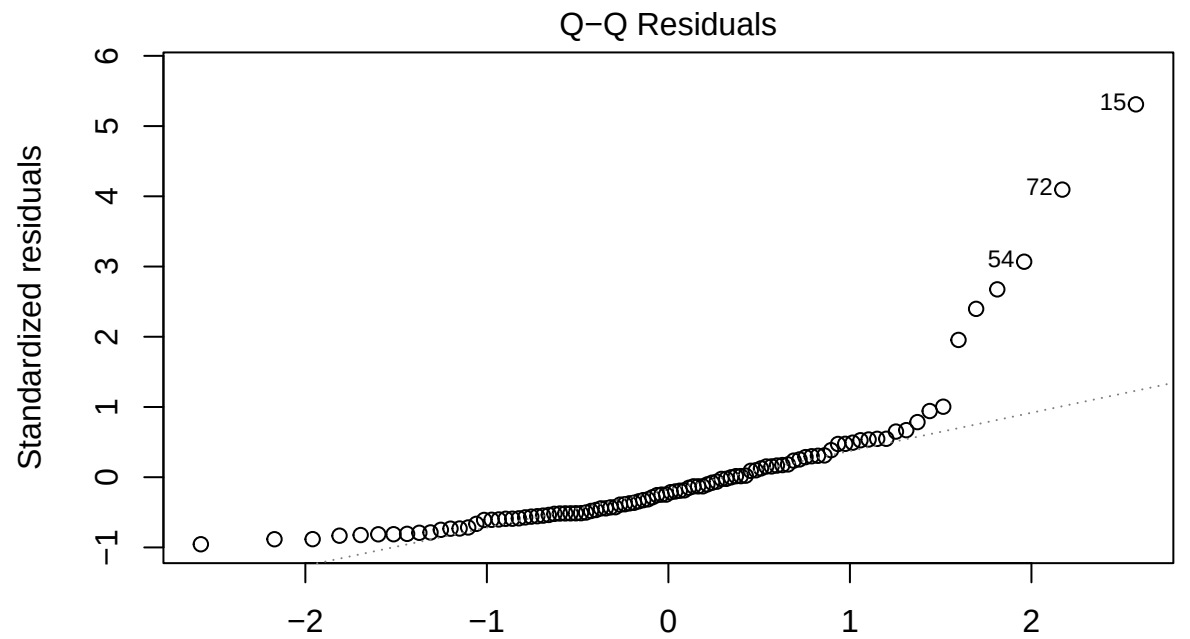
```
modelo3<- lm(z_unif ~ x_unif + y_unif + I(x_unif^2) + I(y_unif^2) + I(x_unif*y_unif) + I(x_unif^3) + I(y_unif^3) + I(x_unif^2 * y_unif) + I(x_unif * y_unif^2), data = matriz_unif)
summary(modelo3)
```

```
##
## Call:
## lm(formula = z_unif ~ x_unif + y_unif + I(x_unif^2) + I(y_unif^2) +
##      I(x_unif * y_unif) + I(x_unif^3) + I(y_unif^3) + I(x_unif^2 *
##      y_unif) + I(x_unif * y_unif^2), data = matriz_unif)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.3973 -0.8040 -0.3362  0.2862  7.9689
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      1.73804    0.27603   6.297 1.09e-08 ***
## x_unif           0.25301    0.18261   1.386 0.169316
## y_unif          -0.09988    0.17544  -0.569 0.570545
## I(x_unif^2)      -0.10230    0.03341  -3.062 0.002902 **
## I(y_unif^2)      -0.12175    0.03422  -3.558 0.000598 ***
## I(x_unif * y_unif)  0.02340    0.03022   0.774 0.440929
```



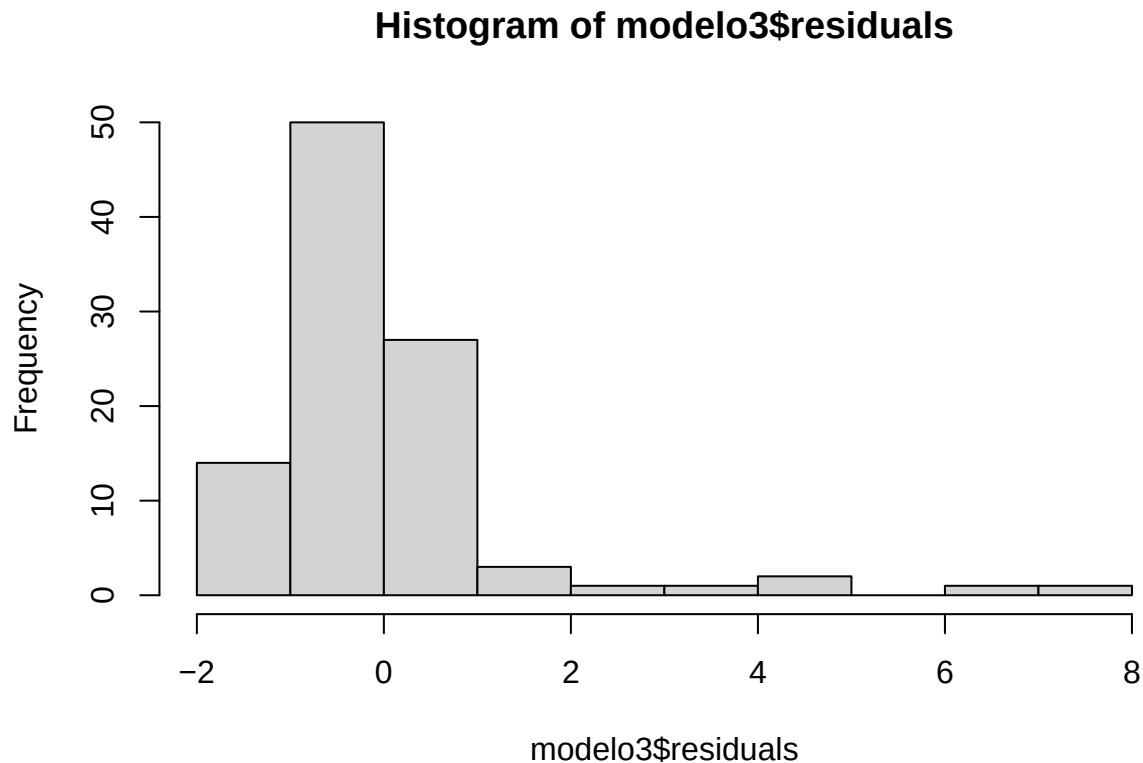
```
## I(x_unif^3)      -0.01638    0.01611   -1.017  0.311967
## I(y_unif^3)      0.01012    0.01564    0.647  0.519286
## I(x_unif^2 * y_unif) 0.00931    0.01404    0.663  0.508863
## I(x_unif * y_unif^2) -0.01164    0.01520   -0.766  0.445810
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.526 on 90 degrees of freedom
## Multiple R-squared:  0.2257, Adjusted R-squared:  0.1483
## F-statistic: 2.915 on 9 and 90 DF,  p-value: 0.004481
plot(modelo3, which = 1:2)
```





lm(z_unif ~ x_unif + y_unif + I(x_unif^2) + I(y_unif^2) + I(x_unif * y_unif ...

```
hist(modelo3$residuals)
```



Observações sobre esses modelos:

residuo = valor_real - valor_previsto

residuo = - valor_previsto (quando o real é zero)

- A maioria desses modelos não atendem às suposições necessárias para o teorema de gauss-markov
 - Os erros são heterocedásticos e correlacionados.
- O modelo prevê valores menores que zero (fora do contradomínio de f)

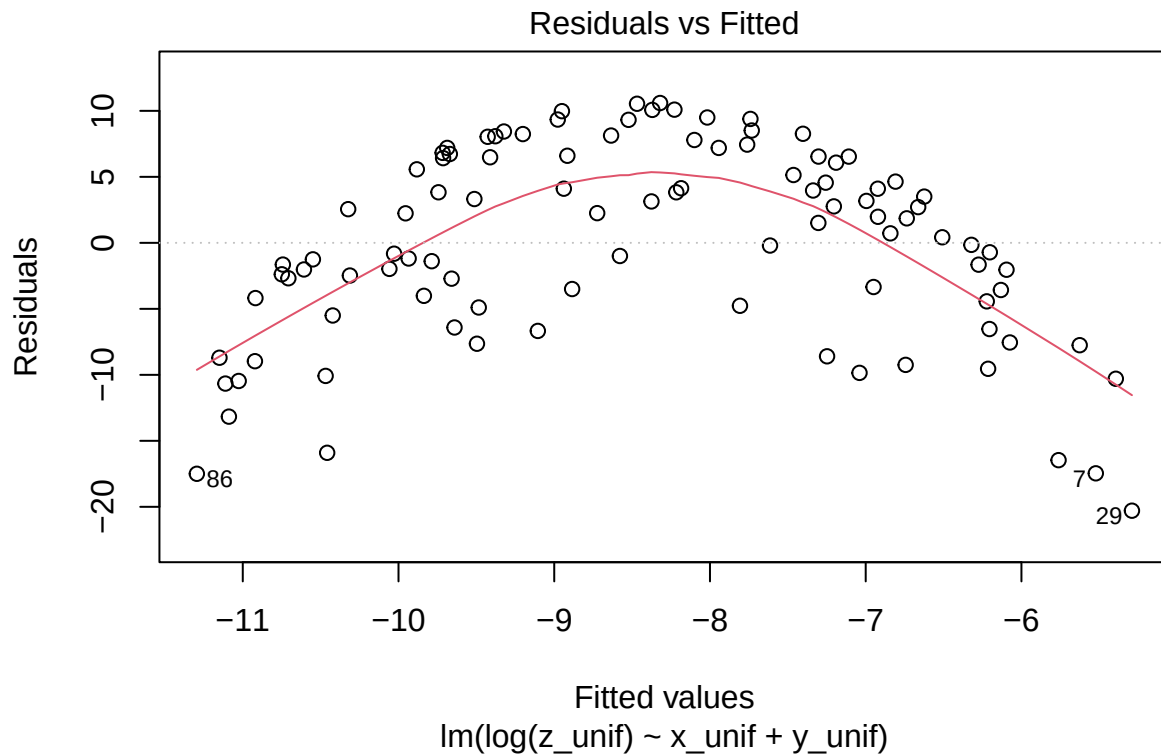
Transformação LN(Y)

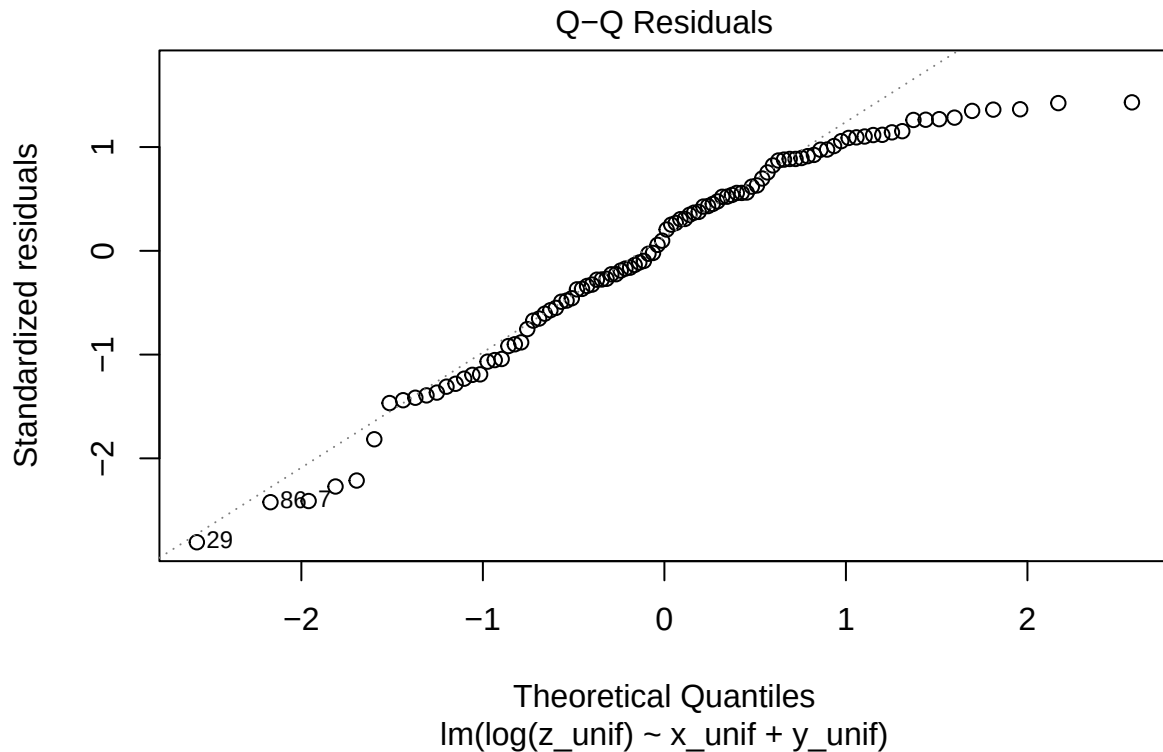
Modelo 1 [$\ln(m1(x,y)) = b_0 + b_1x + b_2y$]

```
modelo1_ln <- lm(log(z_unif) ~ x_unif + y_unif, data=matriz_unif)
summary(modelo1_ln)
```

```
##
## Call:
## lm(formula = log(z_unif) ~ x_unif + y_unif, data = matriz_unif)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -20.291  -4.517   1.113   6.495  10.593
##
```

```
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -8.2053     0.7562 -10.851  <2e-16 ***
## x_unif       -0.1111     0.3335  -0.333   0.7397
## y_unif       -0.6716     0.3247  -2.069   0.0412 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.441 on 97 degrees of freedom
## Multiple R-squared:  0.04712,    Adjusted R-squared:  0.02747
## F-statistic: 2.398 on 2 and 97 DF,  p-value: 0.09623
plot(modelol_ln, which = 1:2)
```





Modelo 2 [$\ln(m2(x,y)) = b0 + b1*x + b2*y + b3*x^2 + b4*y^2 + b5*x*y$]

```
modelo2_ln <- lm(log(z_unif) ~ x_unif + y_unif + I(x_unif^2) + I(y_unif^2) + I(x_unif*y_unif), data=matriz_unif)
summary(modelo2_ln)
```

```
## Warning in summary.lm(modelo2_ln): essentially perfect fit: summary may be
## unreliable
```

```
##
```

```
## Call:
```

```
## lm(formula = log(z_unif) ~ x_unif + y_unif + I(x_unif^2) + I(y_unif^2) +
##      I(x_unif * y_unif), data = matriz_unif)
```

```
##
```

```
## Residuals:
```

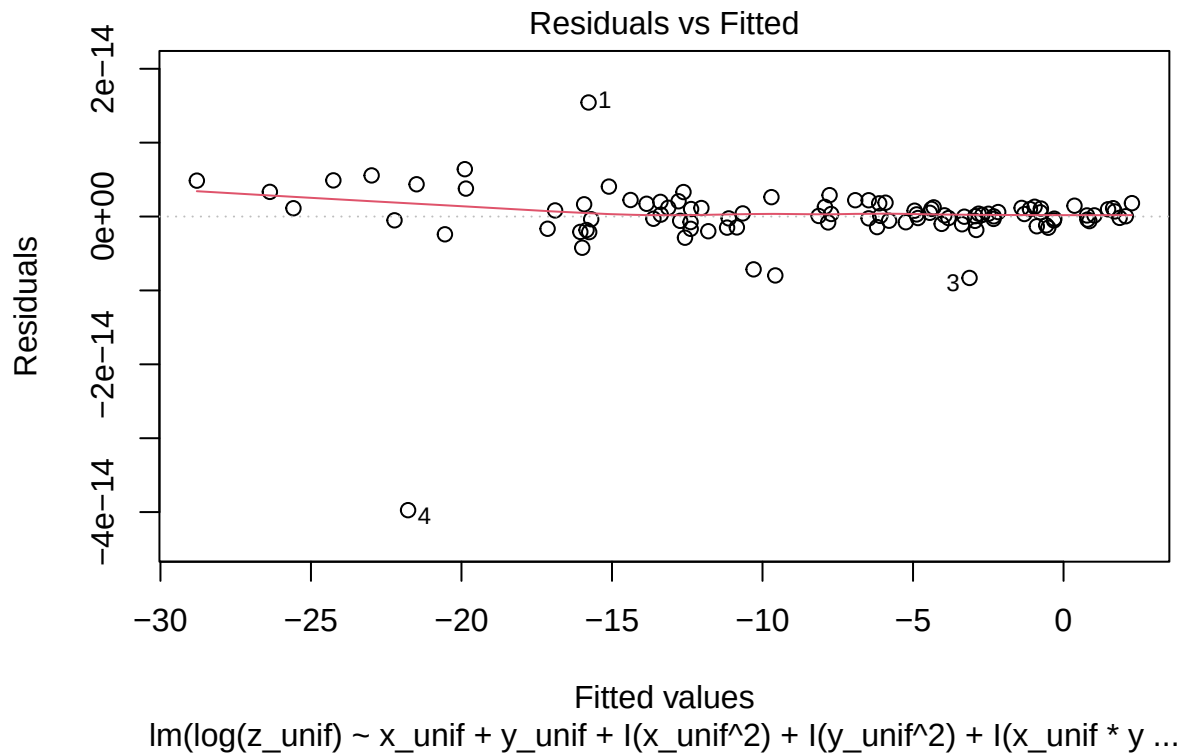
```
##      Min      1Q   Median      3Q      Max
## -3.974e-14 -6.530e-16  1.930e-16  1.257e-15  1.544e-14
```

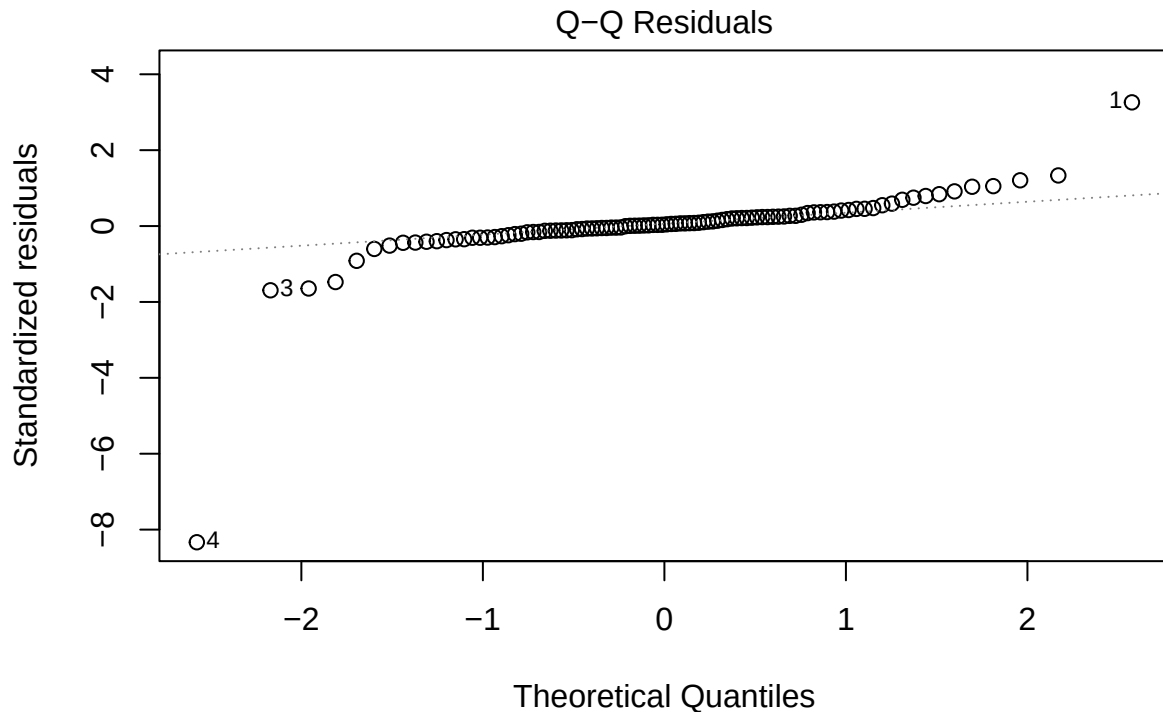
```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error  t value Pr(>|t|)
## (Intercept)    2.303e+00  8.824e-16  2.609e+15  <2e-16 ***
## x_unif         -4.204e-17  2.312e-16 -1.820e-01  0.8561
## y_unif          5.425e-16  2.246e-16  2.416e+00  0.0176 *
## I(x_unif^2)    -1.000e+00  1.077e-16 -9.288e+15  <2e-16 ***
## I(y_unif^2)    -1.000e+00  1.080e-16 -9.256e+15  <2e-16 ***
```

```
## I(x_unif * y_unif) -1.598e-16  9.297e-17 -1.719e+00  0.0889 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.994e-15 on 94 degrees of freedom
## Multiple R-squared:      1, Adjusted R-squared:      1
## F-statistic: 4.52e+31 on 5 and 94 DF,  p-value: < 2.2e-16
plot(modelo2_ln, which = 1:2)
```





$\text{lm}(\log(z_{\text{unif}}) \sim x_{\text{unif}} + y_{\text{unif}} + I(x_{\text{unif}}^2) + I(y_{\text{unif}}^2) + I(x_{\text{unif}} * y_{\text{unif}} +$

Modelo 3 [$\ln(m2(x,y)) = b_0 + b_1*x + b_2*y + b_3*x^2 + b_4*y^2 + b_5*x*y + b_6 *x^3 + b_7 *y^3$]

```
modelo3_ln <- lm(log(z_unif) ~ x_unif + y_unif + I(x_unif^2) + I(y_unif^2) + I(x_unif*y_unif) + I(x_unif^3) + I(y_unif^3), data = matriz_unif)
summary(modelo3_ln)
```

```
## Warning in summary.lm(modelo3_ln): essentially perfect fit: summary may be
## unreliable
```

```
##
```

```
## Call:
```

```
## lm(formula = log(z_unif) ~ x_unif + y_unif + I(x_unif^2) + I(y_unif^2) +
##     I(x_unif * y_unif) + I(x_unif^3) + I(y_unif^3) + I(x_unif^2 *
##     y_unif) + I(x_unif * y_unif^2), data = matriz_unif)
```

```
##
```

```
## Residuals:
```

```
##      Min       1Q   Median       3Q      Max
## -3.850e-14 -7.770e-16  1.840e-16  1.279e-15  1.506e-14
```

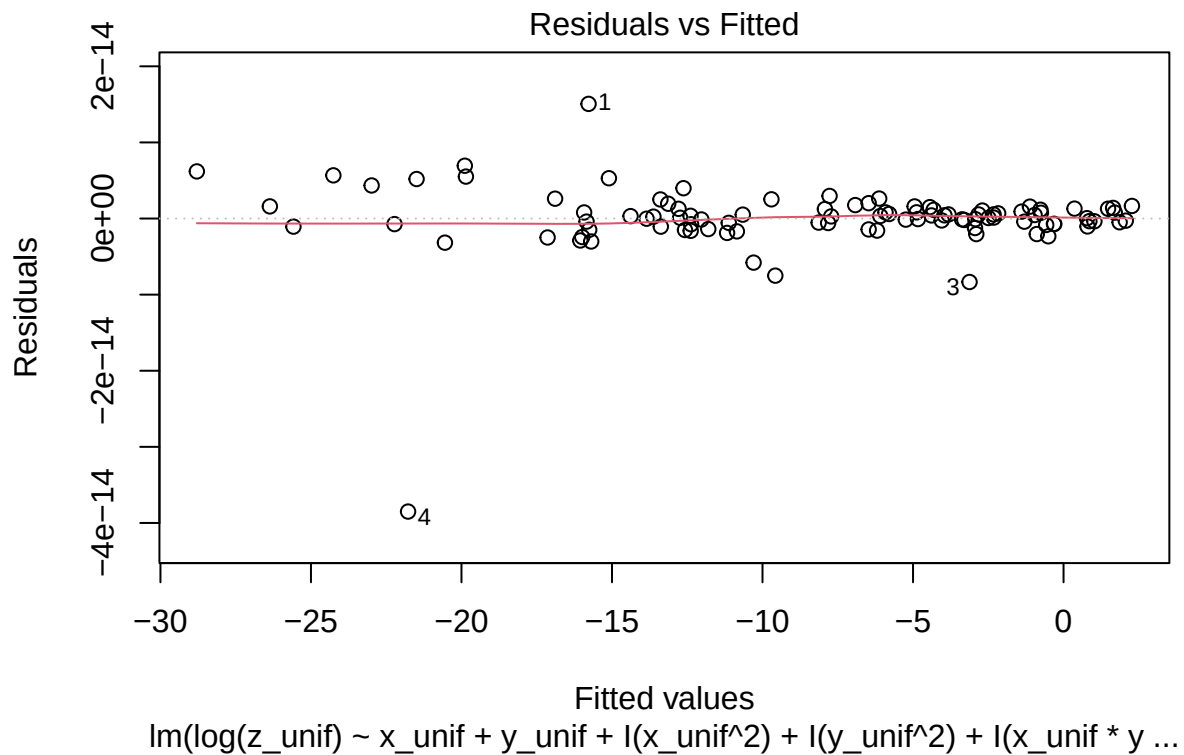
```
##
```

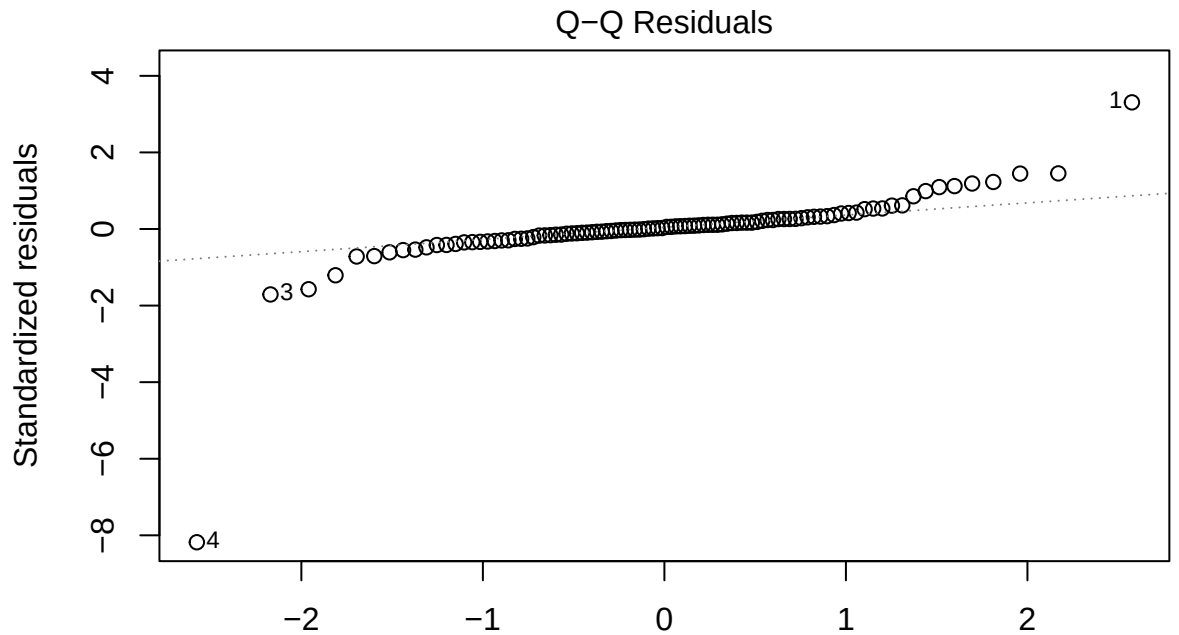
```
## Coefficients:
```

```
##              Estimate Std. Error  t value Pr(>|t|)
## (Intercept)    2.303e+00  9.081e-16  2.536e+15  <2e-16 ***
## x_unif         3.310e-16  6.008e-16  5.510e-01   0.583
## y_unif         8.525e-16  5.772e-16  1.477e+00   0.143
## I(x_unif^2)    -1.000e+00  1.099e-16 -9.097e+15  <2e-16 ***
```

```
## I(y_unif^2)          -1.000e+00  1.126e-16 -8.884e+15  <2e-16 ***
## I(x_unif * y_unif)  -1.075e-16  9.944e-17 -1.081e+00   0.283
## I(x_unif^3)         -4.193e-18  5.298e-17 -7.900e-02   0.937
## I(y_unif^3)         -4.456e-17  5.144e-17 -8.660e-01   0.389
## I(x_unif^2 * y_unif) 3.406e-17  4.618e-17  7.380e-01   0.463
## I(x_unif * y_unif^2) -6.284e-17  4.999e-17 -1.257e+00   0.212
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.02e-15 on 90 degrees of freedom
## Multiple R-squared:  1, Adjusted R-squared:  1
## F-statistic: 2.485e+31 on 9 and 90 DF, p-value: < 2.2e-16
```

```
plot(modelo3_ln, which = 1:2)
```





Theoretical Quantiles
 $\ln(\log(z_{\text{unif}}) \sim x_{\text{unif}} + y_{\text{unif}} + l(x_{\text{unif}}^2) + l(y_{\text{unif}}^2) + l(x_{\text{unif}} * y \dots$

Observações

$$\ln(k \cdot \exp[-c(x^2 + y^2)]) = \ln(k) - cx^2 - cy^2$$

ou seja,

$$k = \exp[b_0]$$

$$c = -b_1 = -b_2$$

- Os termos de grau 3 não foram estatisticamente significativos, como esperado.
- As suposições do modelo foram atendidas e os parâmetros possuem baixo erro padrão;

Logo, o melhor modelo foi:

$$\ln(y) = b_0 + b_1x^2 + b_2y^2$$

$$y = \exp[b_0 + b_1x^2 + b_2y^2]$$